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C204

Midterm Lab Task 2 Using Functions

Problem 1.

Create an n x n Multiplication table using **Nested FOR Loop.** The user must enter the number of rows and columns that will be displayed in the Table.

```
rows = int(input("How many rows: "))
cols = int(input("How many columns:"))

print("\nMultiplication Table")
for i in range(1, rows + 1):
    for j in range(1, cols + 1):
        print(i * j, end="\t")
    print()
```

Sample Output 1

```
How many rows:10
How many cols:10
                        Multiplication Table

   1    2    3    4    5    6    7    8    9   10

   2    4    6    8   10   12   14   16   18   20

   3    6    9   12   15   18   21   24   27   30

   4    8   12   16   20   24   28   32   36   40

   5   10   15   20   25   30   35   40   45   50

   6   12   18   24   30   36   42   48   54   60

   7   14   21   28   35   42   49   56   63   70

   8   16   24   32   40   48   56   64   72   80

   9   18   27   36   45   54   63   72   81   90

  10   20   30   40   50   60   70   80   90  100
```

Sample Output 2.

```
How many rows:3
How many cols:5
                        Multiplication Table

   1    2    3    4    5

   2    4    6    8   10

   3    6    9   12   15
```

Problem 2. Create a bank program that will allow the user to perform the ff: Use Functions as necessary

```python
balance = 0.0

def show_balance():
    print("************************")
    print(f"Your balance is ${balance:.2f}")
    print("************************")

def deposit():
    global balance
    amount = float(input("Enter an amount to be deposited: "))
    if amount > 0:
        balance += amount
        print("************************")
        print(f"${amount:.2f} deposited successfully!")
        print("************************")
    else:
        print("Invalid deposit amount!")

def withdraw():
    global balance
    amount = float(input("Enter amount to be withdrawn: "))
    if amount > 0:
        if amount <= balance:
            balance -= amount
            print("************************")
            print(f"${amount:.2f} withdrawn successfully!")
            print("************************")
        else:
            print("Insufficient balance!")
    else:
        print("Invalid withdrawal amount!")

def main():
    while True:
        print("\n************************")
        print("        ABCDE ATM       ")
        print("************************")
        print("1. Show Balance")
        print("2. Deposit")
        print("3. Withdraw")
        print("4. Exit")
        print("************************")

        choice = input("Enter your choice (1-4): ")

        if choice == '1':
            show_balance()
        elif choice == '2':
            deposit()
        elif choice == '3':
            withdraw()
        elif choice == '4':
            print("Thank you for using ABCDE ATM. Goodbye!")
            break
        else:
```

```
            print("Invalid choice! Please enter 1-4.")

main()
```

Sample output:

```
*********************
      ABCCDE ATM
*********************
1.Show Balance
2.Deposit
3.Withdraw
4.Exit
*********************
```

```
Enter your choice (1-4): 1
*********************
Your balance is $0.00
*********************
```

```
*********************
Enter your choice (1-4): 2
*********************
Enter an amount to be deposited: 1000
*********************
```

```
Enter your choice (1-4): 1
*********************
Your balance is $1000.00
*********************
```

```
Enter your choice (1-4): 3
*********************
Enter amount to be withdrawn: 250
*********************
```

```
*********************
Enter your choice (1-4): 1
*********************
Your balance is $750.00
*********************
```